The Idea Gap: Why Open LLM Agents Fail New Environments

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Abstract

When a language-model agent fails an unseen task, the score does not say what was missing: context, tools, time, a world model, a larger model, or the idea. We measure it on the agent's own failures: on ARC-AGI-3, where an agent must learn an unseen game from its actions, we replay 35 failed level attempts of the strongest open agent, restore its conversation and code, and continue under one change at a time against repeated runs of the unchanged agent. Clears with the level's key idea supplied minus clears without it is the agent's idea gap, corrected with a same-length placebo. Rules written from the game source and checked on the engine show the mechanics right and the idea missing in 22 of 35 failures. Context handling, repairs, prompting, reasoning effort, a cross-game library and offered world-model tools stay inside the unchanged range; a larger model of the same family falls below it. Two levers move clears in every setting, and they add: time, and the idea. Told it, the agent clears 116 of 157 public levels against 90 and its full-game score rises from 48.96 to 66.06; a same-length block of true but irrelevant rules recovers part of the gain. Ideas the agent writes for itself lower clears (76 against 90). A copy of the real engine as a simulator clears 9 of 22 idea-missing levels alone and 16 with the idea. We release the rules, cases, test set and protocol as a measure for any agent.

1 Introduction

Benchmarks for agents report one number per task. When the task is one the agent has never seen, the number says that it failed and not why. A failure could be the agent's memory of what it has tried, the tools it was given, the time it had, its model of how the environment works, the size of its network, or the one idea the task turns on. Each of these is the premise of a line of work, and each is bought with a different budget. Which one is binding is an empirical question, and single whole-task runs cannot answer it: on the benchmark we study, a game's score has a median standard deviation of 16 points over four passes of the same notebook, and two identical submissions of one ARC-AGI-2 notebook read 34.31 and 29.31.

We answer it on the agent's own failures. The setting is ARC-AGI-3, an interactive benchmark of grid games whose rules are unseen and must be learned from the agent's own actions; the agent is the strongest public open one, a harness around Qwen3.8-Flash-Next that reaches 48.96 on the 25 public games and 25.6 to 30.3 on the hidden board. We replay 35 of its failed level attempts to a mid-level state on the real engine, restore its conversation and the functions it wrote, and continue under one change at a time; one such condition, run on every case, is an *arm*. Arms are read against repeated continuations of the unchanged agent (the floor), a placebo text of the same length as the help given, and the level's key idea told (the ceiling); rules written from the game source and checked on the engine make "did the agent find the idea" a checkable label rather than a judgement.

* Null Hypothesis Enthusiasts is an independent research collective for unusually curious people to pursue important questions with people they might never otherwise meet. The rules, case bundles, test set and code of this paper are released under its name (Ethical considerations).

The difference between clears with the idea told and clears without it is the agent's *idea gap*. It is a number any agent can be given on this fixture (the replayed states and their arms), and it separates two things a score conflates: what the agent knows about the environment, and whether it finds the idea that the task turns on.

Five questions organise the paper.

1. **RQ1. When the agent fails a level, is it missing the game's mechanics or the level's idea?** (Section 4.1.) In 22 of 35 failures it stated the mechanics correctly and never stated the level's key use of them; in 2 the mechanics were wrong. Levels where it had found the idea are cleared by continuations; levels where it had only the mechanics (the *mechanics-only* levels) mostly are not, whatever the arm.
2. **RQ2. Does telling the agent the idea raise clears, does a same-length placebo text do the same, and does anything else move them?** (Section 4.) Told, the agent roughly doubles clears on the mechanics-only levels, and the gain carries to clean starts on 157 public levels (116 against 90) and to full games (66.06 against 48.96). The placebo recovers part of the gain; the idea's margin over it passes its pre-set size and sits just above its pre-set p, a reading of report. Everything else tried stays inside the unchanged range, gives no decision, or falls below it, except exposing the game's restart action, which moved clears on the stuck states at the longer budgets, about half of its gain being time, and not in full games (Table 1).
3. **RQ3. Do more time, or ideas the agent writes for itself, raise clears without the idea being told?** (Section 5.) Time roughly doubles clears when doubled, and adds to the idea rather than replacing it: a supplied idea still needs about twice the budget to act on. Ideas the agent writes for itself lower clears (76 against 90), and ideas from its proposals, a tuned writer or a library do not raise them.
4. **RQ4. Is a world model, given or learned, enough?** (Section 6.) A copy of the real engine as a simulator clears 9 of the 22 mechanics-only levels, 16 with the idea, the idea alone 12; the model helps more once the idea is known. Bounded search on the real engine finds plans on 35 of 183 levels at the largest node budget tried, a lower bound under its wall caps and still about one level in five. Models the agent writes are rarely built or verified, and laws fitted to transitions score well by predicting that nothing changes.
5. **RQ5. Are these levels hard for people, and does the gap hold beyond this model?** (Sections 4.2, 5.1 and 7.) Not for people: those who reach these levels clear them at a median rate of 1.00. The told idea raises clears on a second checkpoint of the same model too (24 against 17), and on ARC-AGI-2 the pattern repeats in the static case: the right output is never generated for 50 of 65 outputs, and a perfect checker does not substitute for generation. Agents on other model families are the open part of RQ5 (Section 9).

The contributions are:

- **A measure.** The idea gap: clears with the level's idea told minus clears unchanged, from an agent's own replayed failure states, with repeated unchanged samples as the floor and a same-length placebo as the control; and a fixture on which any agent can be given the number: 35 restored states, a 22-level test set, 157 level starts and 794 engine-checked rule statements, released with the replay, restore and analysis code.
- **Readings for the strongest open agent.** On its failures the mechanics are right and the idea missing (22 of 35); time and the told idea are the levers that move clears in every setting, and they add; a same-length placebo recovers part of the idea's gain; a perfect world model is worth little before the idea and more after it; ideas the agent writes for itself lower clears.
- **Method findings that travel.** A truncated long run does not measure a short budget; a fitted world model scored by next-step match is rewarded for predicting stasis; and a claim that guidance helped an agent needs a placebo arm.
- **A build order** for agents of this kind, and the benchmark's three open problems answered for one agent (Section 8).

2 Related work

ARC and its human baselines. ARC-AGI-3 measures skill acquisition in interactive environments (Chollet, 2019): an agent plays games it has never seen, with no instructions, and is scored against the action counts of human play (ARC Prize Foundation, 2026). People solve most ARC tasks: in H-ARC, 790 of the 800 public tasks of the original corpus were solved by at least one of 1,729 participants within three attempts (LeGris et al., 2025). A think-aloud dataset of people learning the public ARC-AGI-3 games marks when each player recognises each mechanism (Liu et al., 2026); Section 7 aligns its recognition moments with the levels the agent fails. The agent measured here is the public notebook of the best open harness at the competition's second milestone (Franzen, 2026), derived from the Duck harness (Tufa Labs, 2026).

Evaluating stochastic agents. Reinforcement-learning evaluation has documented how seed variance and small numbers of runs make reported improvements unreliable, and argued for interval estimates and reports of variability (Henderson et al., 2018; Agarwal et al., 2021). The repeated unchanged samples of Section 3 apply the same discipline to a language-model agent whose whole-game run costs about eight GPU-hours. Paired binary outcomes are read with McNemar's exact test (McNemar, 1947), labeller agreement with Cohen's kappa (Cohen, 1960), whole-game differences with the percentile bootstrap (Efron and Tibshirani, 1993), and every run after the first under a rule written before it ran, in the sense of pre-registration (Nosek et al., 2018).

Returning to states. Go-Explore exploits deterministic simulators by returning to saved states and exploring from there (Ecoffet et al., 2021). The measure uses the same property for measurement rather than search: ARC-AGI-3 games are deterministic, so a recorded action prefix reconstructs the board exactly and the agent's state can be rebuilt around it, which is the logic of a counterfactual comparison (Pearl, 2009). Re-running a language-model agent from an intermediate step of a failed trajectory has been used to locate errors: AgentDebug defines the critical step as the earliest whose correction would prevent the failure and re-runs the agent from that step with feedback (Zhu et al., 2025). Here the substitution is the level's idea, and the comparison is against the unchanged agent's own spread.

Failure analysis of language-model agents. Taxonomies of agent failures built by annotating trajectories are an established method: MAST for multi-agent systems, from expert-annotated traces (Cemri et al., 2025), and an error taxonomy for single agents on ALFWorld, GAIA and WebShop (Zhu et al., 2025). The labels of Section 4 are paired with interventions on the same cases and with ground truth extracted from the environment's source, so that a label can be checked against the actual rule and against what changes the outcome.

World models. Learning the environment's dynamics and planning in the learned model is the leading proposal for agents in unseen environments (Ha and Schmidhuber, 2018; Hafner et al., 2023; LeCun, 2022). Section 6 gives the agent a model in four forms, from a copy of the real engine to laws fitted to its own observations, and asks what each buys an agent that already has the mechanics.

Guidance in context. What a language model does with the same weights changes with what it is given in context: reflection text and interleaved reasoning change an agent's later actions (Shinn et al., 2023; Yao et al., 2023), and prompt and exemplar design moved subjectivity labels in low-resource languages and the gender balance of generations (Dwivedi et al., 2023; Dwivedi et al., 2024). The told-idea arm, read against a same-length placebo, asks the narrower question of whether a text's content, not its presence, changes what an agent clears. The failures resemble a known gap between choosing experiments that discriminate between hypotheses and testing a favoured one (Lindley, 1956; Lake et al., 2017); the probe-first, repertoire-sweep and self-proposed arms test remedies for that gap, and the told idea bounds how much the right hypothesis would close.

The competition's own entries. The entry that led the hidden board at 75.56 reports the same diagnosis from its own tests on the public games: its base model was about 20 to 23 points short of a perfect score, 18 to 20 of them from stages it could not solve at all; both of its interventions act on stuck states, a read-only reviewer after 45 minutes on one stage and reinforcement learning from fresh and stuck snapshots, and no test isolating the learning's gain is reported (Chen, 2026). The sixth entry reports that late in a run most sessions sit stuck on hard levels and the level rate flattens (Zatravkin, 2026). Both figures are the authors' own reports.

3 The measure

3.1 Setting and agent

ARC-AGI-3 (Chollet, 2019; ARC Prize Foundation, 2026) presents grid games with hidden rules; an agent sees frames, chooses actions and must clear each level to reach the next. The competition set holds about 110 hidden games, half of them on the public leaderboard, played in one offline run of at most 9 hours on one 96 GB GPU. Twenty-five public games with 183 levels are available for development, with their engine source.

The agent, P0, is the unmodified public notebook of the best open harness (Franzen, 2026), derived from the Duck harness (Tufa Labs, 2026) and run as published: Qwen3.8-Flash-Next in a W4A16 quantisation with a draft head, 10 concurrent game streams under a priority scheduler, a 131,072-token context with trimming, and the model's own Python functions retained across turns. On the public games it scores 48.96 (25 games, 4 passes); on the leaderboard three submissions of the same notebook read 27.01, 25.62 and 27.94, and a later FP8 build, the entry this paper attaches, 27.54 and 30.26. Across six public-game runs of P0 and variants, a 4-pass run mean has a standard deviation of about 2.2 points and a single game's four passes a median SD of 16.4, so a whole-game difference of about 10 points is needed for 80% power. That instrument cannot read the levers below.

3.2 Replay, restore, compare

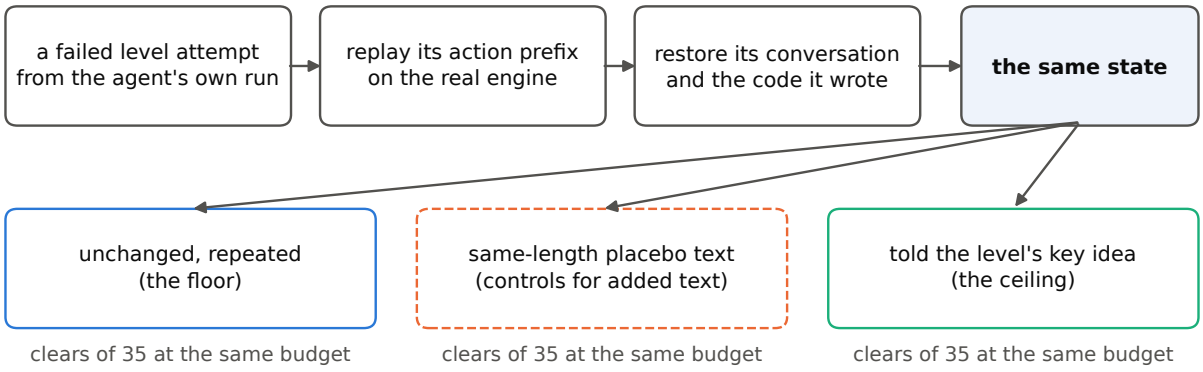


Figure 1: The measure. A failed level attempt is replayed on the real engine to the cut, the agent's conversation and code are restored, and the same state is continued under each arm at the same budget: unchanged (repeated, the floor), a same-length placebo text, and the level's key idea told (the ceiling). The idea gap is told minus unchanged; the placebo controls for added text.

Cases. From P0's public-game run we drew, with a fixed seed, 35 failed level attempts: 10 in the five games that score below 20 in every run and all 25 failed attempts at level 6 or later elsewhere. A failed attempt is the level a pass was on when it ended without winning. The 35 fall on 25 distinct (game, level) pairs in 19 games.

Cut and replay. The cut is placed half-way through the actions P0 spent on the failed level, then moved back to the last turn boundary, so the board and the restored conversation agree (median 26 replayed actions on the level, range 0 to 69). The prefix, resets included, is replayed through the harness's own action path with no generated tokens, so the runtime state, the history shown to the model and the event log come out as in P0. Before any arm ran, every prefix was replayed and compared board by board with the recorded events: 35 of 35 match locally and 35 of 35 on the competition's own wheels.

Restore. The conversation of the last turn before the cut is reinstated from P0's request log (median 59 messages), and the latest definition of every function the model wrote before the cut is re-executed once so the harness re-adopts it (median 21 functions). State outside the messages is not restored. Every arm shares these approximations, so comparisons are between arms, never against P0's own outcome.

Budget and outcome. The standard budget is 80 actions or 15 minutes after the restore, whichever comes first; doubled, 160 actions or 30 minutes; quadrupled, 320 or 60. A case under one condition is a *case-arm*; it stops at the clearing action. The outcome is whether the failed level is cleared. The model sees the remaining wall time; the action cap is enforced silently.

Reading. An arm's *reading* is its clears set against the unchanged samples at the same budget and, where one was written, its decision rule. Each arm is paired with the unchanged arm by case: discordant counts and the exact McNemar test (McNemar, 1947). The unchanged arm is itself sampled repeatedly, five times at the standard budget and four at the doubled one, so that an arm is read against the floor's spread and not against one draw. With 35 pairs, an arm that clears a superset of the unchanged levels is detected at 80% power only if it adds about 22 percentage points: the design detects large effects, and a reading inside the unchanged range is a bound, not a null. Every run after the first had its arms and its decision rule written down before it ran, from the fifth on as a frozen rule file.

3.3 Floor, placebo, ceiling: the idea gap

Three reference arms turn a comparison into a measure.

- The **floor** is the unchanged agent, sampled repeatedly at the same budget: 9, 11, 6, 5 and 7 of 35 at the standard budget; 14, 14, 18 and 17 at the doubled budget. Two samples agree on 29 of 35 levels, so outcomes from a restored state are largely reproducible at the level scale, and an arm whose discordance with the floor exceeds the floor's own (6) changed which levels clear.
- The **ceiling** tells the agent the level's key idea in one message at the restore point: the key-insight statement from the rules of Section 3.4. Anything the agent cannot do with the idea in hand is not for want of it.
- The **placebo** is a block of true core rules of the same game, chosen to share no content word with the level's idea, matched to its length within 20%, in the same message position. It separates the content of the help from the presence of a rule-shaped text.

The idea gap is the ceiling's clears minus the floor's mean, with the placebo's margin over the floor reported beside it. For this agent at the doubled budget on the 35 states the figures are 21.5, 15.75 and 18 (Section 4). The same three arms run from clean level starts give the gap on the whole public set (Section 4.3). The fixture and the three arms are what another agent needs to be measured the same way.

3.4 Ground truth from the game source

For each public game we wrote the rules from the game's source: what each action does, the core rules, and per level the layout, what is new, the key insight and a winning strategy. Each statement ends with its source line range, a claim id and a status. Where a statement has a mechanical consequence it is checked on the engine: the check reaches the level, runs an action sequence and judges a recorded expectation (cleared on the last action and not before; not cleared; game over). Of 794 statements, 629 are engine-checked and 165 source-only; re-running every check gives 547 pass, 82 marked for inspection and 0 failures. For every failed (game, level) pair both the key insight and the win condition are engine-checked, usually with a contrast showing that the naive alternative does not clear. Rules were written before the failure dossiers were read, and no key insight was changed afterwards; a probe tool re-runs any check with per-step board diffs, so a disputed statement is settled on the engine.

Against these rules each failed attempt is labelled by whether the agent discovered the level's key insight: *yes* if it stated it as its working belief at some turn, *partial* if its core mechanics were right and the insight missing or half-stated, *no* if the mechanics were wrong. Every evidence quote is checked verbatim against the dossier by script.

3.5 Preconditions

The measure needs a deterministic engine that can replay a recorded prefix to the same state, an agent whose context and code can be reinstated, a budget that can be fixed per case, and a statement of each task's idea that can be checked against the engine. Grid games, text adventures, program-synthesis tasks with checkers and most simulated control tasks meet the first three; the fourth is work, and it is the work that turns a label into ground truth. Section 7

applies the floor and the stable-subset reading (which levels every arm clears and which none does, against independent draws) to ARC-AGI-2, where the idea is the transformation and the checker is the demonstration pairs.

3.6 Hypotheses, and what was fixed in advance

Every continuation run after the first had its arms and its decision rule written down before it ran, from the fifth on as a frozen rule file (Appendix D), in the sense of pre-registration (Nosek et al., 2018); a reading is stated as met, not met or report, and no threshold was changed after a result. The pre-registered hypotheses, among them:

- **RQ2.** Told the idea at the standard budget, the agent clears 8 or more of the 22 mechanics-only levels: not met (6). At the doubled budget the told arm's repeat clears 10 or more of 22: met (12 and 12). At four times the budget it clears 26 or more of 35: met (29). From level starts the idea generalises at a gain of 10 levels with p at most 0.01: met (26, p 0.0001); its margin over the placebo supports content at 10 with p at most 0.01: report (16, p 0.011). The placebo at the doubled budget supports content at 17 or fewer clears and rejects it at 20 or more: report (18). On the second checkpoint the told idea replicates at 21 or more: met (24).
- **RQ3.** The self-written idea helps at a gain of 10 with p at most 0.01: not met (76 against 90). A within-run control predicted that the doubled-budget arm's clears within the standard budget would sit inside the floor: failed (Section 5.1).
- **RQ4.** The enforced world-model tools are promoted at 22 of 35 or 11 of 22: report (16 and 8). The fitted law library's rule gives no decision, since the library was rarely called (Section 6.4).
- **RQ5.** Arms on a second model family read the same shape if the idea gap is at least 6 of 35 with p at most 0.05 and the unchanged arm clears at most 25, no gap if the gap is at most 2 with the unchanged arm at most 25, and closed by capability if the unchanged arm clears 26 or more; fixed before any such arm ran. The human second rating was pre-registered before any kit was sent (Appendix C); the think-aloud readings were pre-set as descriptive, with no decision threshold; the ARC-AGI-2 rules are in Appendix F.

Exploratory, tied to no written rule: the first continuation run; the rule-discovery labels of Section 4.1, observed after the runs by a labeller not blind to the outcomes and tested by intervention in Section 4.2; comparisons pooled across runs; pass-level tests in whole-game runs; the stable-subset reading across arms; the public-play clear rates of Section 7.1; the efficiency accounting of Section 8.3; and the mechanism in the fitted laws of Section 6.4.

4 Mechanics right, idea missing

RQ1 asks whether a failure is missing the mechanics or the idea; RQ2, whether the idea, told, raises clears, whether a same-length placebo does the same, and whether anything else moves them.

4.1 The label and what it predicts

Two labellers, the second blind to the first, read a dossier of each failure (the level's budget, every action with its board change, every model turn with a short quote) and agreed on a primary failure category in 23 of 35 (Cohen's kappa 0.54; Cohen, 1960); both found a cheap, discriminating next experiment visibly available in 34 of 35 failures. Against the engine-checked rules, the rule-discovery labels for the 35 attempts are *yes* 11, *partial* 22, *no* 2. In 22 of 35 failures the agent had the game's mechanics right and was missing the level's specific use of them, typically a mechanic already in its repertoire that it had ruled out or never tested: blasts as a means of transport, a clickable box as a button-holder, relaying through a helper, a third recording that clears all ghosts. In only 2 were the mechanics wrong. Of the 11 that stated the idea, 10 acted on it; all 11 ran out of time or were paused with the level uncleared, and in one the exact 21-move winning plan had been simulated before the run ended.

The label predicts which levels continuations clear (Table 2). **Table 2. Continuation outcomes by rule-discovery label, 35 failed levels (ARC-3).**

Label	n	cleared by either of two unchanged samples, 15 min	cleared unchanged, 30 min	cleared by any of 8 early arms
yes	11	9	11	11

Label	n	cleared by either of two unchanged samples, 15 min	cleared unchanged, 30 min	cleared by any of 8 early arms
partial	22	4	3	8
no	2	0	0	0

Fisher's exact test, *yes* against *partial*: p 0.0007 at 15 minutes, below 0.0001 at 30. Pooled over three unchanged 15-minute samples, *yes* levels cleared 17 of 33 times and *partial* levels 9 of 66 (p 0.0002); each of five unchanged samples cleared 3 of the 22 *partial* levels. In 8 of the 11 *yes* cases P0 first stated the idea after the cut, so the restored conversation did not contain it: the label marks levels whose idea this agent can reach from that state, not information carried in its context. The label is observed, not assigned, and the labeller was not blind to continuation outcomes; Section 4.2 therefore tests it by intervention.

4.2 Told the idea

At the standard budget the idea did little: 12 of 35 and 6 of 22 against a same-run unchanged sample of 5 and 3, with the placebo at 10 and 4 and the win condition alone at 7 and 2. At the doubled budget, told the idea in one message at the restore point, the agent cleared 20 of 35 and 12 of the 22 mechanics-only levels, and in an independent repeat 23 and again 12, against unchanged samples of 14, 14, 18 and 17 (mechanics-only 3, 7, 8, 8). At four times the budget it cleared 29 of 35 and 17 of 22, the most of any arm, against 21 and 10 for time alone (paired 10 against 2, p 0.039). Time and the idea add: about 15.75 of 35 unchanged at 30 minutes, 21 with four times the time, 21.5 with the idea, 29 with both. On a second checkpoint of the model the idea cleared 24 against 17 (7 levels only with it, none only without, p 0.016).

The placebo at the doubled budget cleared 18 of 35 and 9 of 22, at the top of the unchanged range. The idea's margin over a same-length text is 2 to 5 levels overall and 3 on the mechanics-only levels at 30 minutes, and 2 at 15 (p 0.77): on the stuck states these runs show that told the idea the agent clears more, and that a rule-shaped text recovers part of the gain, not that the idea's content is what moves it. That question is settled, as far as one run settles it, from level starts.

4.3 From level starts, and in full games

On each of the 157 public levels whose key idea carries an engine-verified claim, the agent played from a clean level start, reached by a replayed and hash-checked prefix, for 160 actions or 30 minutes, unchanged and told the level's idea as its first message, both arms in the same run and paired by level under a rule frozen before the run. Unchanged it cleared 90 and told the idea 116: 35 levels cleared only with the idea and 9 only without (exact McNemar p 0.0001), the rule's "generalises". On the 146 levels with a length-matched placebo text, the idea cleared 107, the placebo 91 and unchanged play 81: the idea's margin over the placebo is 16 (26 only with the idea, 10 only with the placebo, p 0.011), which passes the rule's pre-set margin of 10 and sits just above its pre-set p of 0.01, so the rule reads report. The placebo's 10 over unchanged play (p 0.12) recovers about 38% of the gain. The idea also changed how the agent played: on the 146 levels with a placebo, 1,615 tokens per action against 1,996 with the placebo and 2,078 unchanged, 2.84 actions per case-minute against 2.06 and 1.99, and 38 case-arms ending at the wall against 52 and 65. Told the idea the agent thinks less per move and moves more; the placebo does neither.

Shown once at the first model call of each of those 157 levels in full public games, with nothing else changed, the idea raised P0's score from 48.96 to 66.06 (22 of 25 games higher, 2 lower, 1 equal; sign-flip p 0.0001; game-bootstrap interval +10.6 to +24.1, Efron and Tibshirani, 1993). The gain is not confined to replayed states. Its texts come from the game source, which hidden games do not have; the run has no placebo arm; it is one run on public games.

4.4 What did not move clears

Table 1 lists every arm run from the 35 states with its reading against the unchanged samples at the same budget. The pattern is the paper's second result: apart from time, the idea and the restart option (about half of whose gain is time, and none in full games), nothing moved clears outside the unchanged range, and what moved them moved them down.

Table 1. Levels cleared from the same 35 replayed states by each arm, against every unchanged sample at the same budget (ARC-3). Mechanics-only: the 22 *partial* levels.

Lever	Arm	Budget	Cleared of 35	Mechanics-only of 22	Reading
floor	unchanged, five samples	15 min	9, 11, 6, 5, 7	3 in each	
floor	unchanged, four samples	30 min	14, 14, 18, 17	3, 7, 8, 8	
floor	unchanged	60 min	21	10	time compounds
context	fresh start, no conversation or functions	15 min	1		below (p 0.008); context carries progress
context	conversation trimmed to 45%, as P0 trims	15 min	7		inside
context	retained functions dropped	15 min	6		inside
repair	probe-first prompt	15 min	5		inside, 12 discordant
repair	tool-timeout repair	15 min	6		inside; timeouts 38 to 1
prompt	repertoire sweep	15 min	8	4	inside
prompt	rule card (keep a rule, predict, rewrite)	30 min	14	6	inside
effort	reasoning effort low instead of highest	15 min	7	2	inside; lever closed
idea	told the key idea	15 min	12	6	above the range (same-run sample 5; no paired test against it; against the placebo 7/5, p 0.77); pre-set rule not met
idea	told the key idea, two runs	30 min	20, 23	12, 12	above the range; replication rule (10 or more of 22) met at 12 and 12; pooled unpaired p 0.005, indicative
idea	told the key idea	60 min	29	17	above time alone (p 0.039)
placebo	same-length true but irrelevant rules	15 / 30 min	10 / 18	4 / 9	top of the unchanged range
idea, other games	57 patterns from the other games' rules	30 min	16	8	inside (same-run 17, 8)
idea, learned	insight written by a 27B tuned on other games' insights, against the same weights untuned	30 min	13 against 19		no transfer
idea, self-proposed	three uses of seen mechanics proposed by the same model at each failure	30 min	17	6	inside; right mechanic 8 of 15, full key use 0 of 24
restart	the game's RESET action exposed	15 / 30 / 60 min	10 / 20, 21 / 26, 25	3 / 9, 11 / 14, 14	above the range at 30 and 60 (paired p 0.11 to 0.79 at 30, mechanics-only p 0.008 in one run; p 0.18 and 0.125 against time alone at 60); about half of the 60-min gain is time; no gain in full games
model	27B of the same family, low and full effort	30 min	0, 0	0, 0	below; rarely commits to a move
model	second checkpoint, unchanged / told	30 min, 320 actions	17 / 24	7 / 13	inside / above (p 0.016)
world model	engine simulator alone / with the idea (22 levels)	30 min		9 / 16	Section 6

Lever	Arm	Budget	Cleared of 35	Mechanics-only of 22	Reading
world model	write, verify and plan tools, verification enforced	30 min	16	8	report (rule: promote at 22 of 35 or 11 of 22, neither met); inside the unchanged range; verify ok 10 of 90
world model	fitted law library as tools / with proposals tested in it	30 min	20 / 10	10 / 3	no decision (library rarely called); proposals lost 10 cases (p 0.002)

Readings: "inside" means within the unchanged samples' range at that budget, with paired p above 0.05 against each where a test was run; "above" and "below" mean outside that range, with the paired test stated where one was run. Across the thirteen standard-budget arms of the first six runs (four unchanged samples, the probe, repair, context, sweep, idea, win-condition and placebo arms), 23 of 35 levels were cleared by at least one arm against 33.3 expected if arms drew independently at their observed rates, and 12 by none (permutation p 0.019): which levels clear is a property of the level and the agent, and the budget moves the boundary.

Removing all of the context destroyed progress (1 of 35; the arm ran out of time in 34 of 35 case-arms), but removing either half of it cost little (7 and 6 against 7 on the levels both unchanged samples cleared): the conversation and the retained functions are redundant carriers of what the agent has learned, and P0's own trimming is not a material loss. Repairs did what they were built to do and did not change clears. The larger model of the same family, in this harness, analysed in code and rarely moved. The restart option is the one deployable change that moved clears on stuck states; it needs time on the level that a full run does not give, and in full public games it did not help.

4.5 The idea gap for this agent

On the 35 states at the doubled budget: ceiling 21.5, floor 15.75, placebo 18, so the gap is 5.75 of 35 and the placebo-corrected gap 3.5. From level starts on 157 levels: 26 of 157; on the 146 with a placebo, 16 of 146 after correction. In full games: 17.1 points of 100. The answer to RQ1 is the idea, missing in 22 of 35 failures with the mechanics in hand; to RQ2, that the told idea raises clears from stuck states, from clean starts and in full games, that a same-length placebo recovers part of the gain, and that nothing else tried moved clears outside the unchanged range except time (Section 5) and the restart option, about half of whose gain is time. Section 5 asks what it costs to use an idea once given, and Section 6 whether a world model closes the gap.

5 The idea needs budget, and the agent cannot write it

RQ3 asks whether more time, or ideas the agent writes for itself, raise clears without the idea being told.

5.1 Time

Time is the one lever besides the idea, and it is measured against independent samples rather than by truncating long runs. Five unchanged samples at the standard budget cleared 38 of 175 case-arms; three at the doubled budget cleared 46 of 105 (p 0.0001), a fourth 17 of 35; one at four times the budget cleared 21 of 35 and 10 of 22. Doubling the budget on a stuck level roughly doubles clears. The engine-verified routes for the 25 levels of these states need a median of 37 actions (interquartile range 24 to 47), none more than 160, so the doubled budget does not bind on actions; what binds is the agent's own route and the thinking it spends per move.

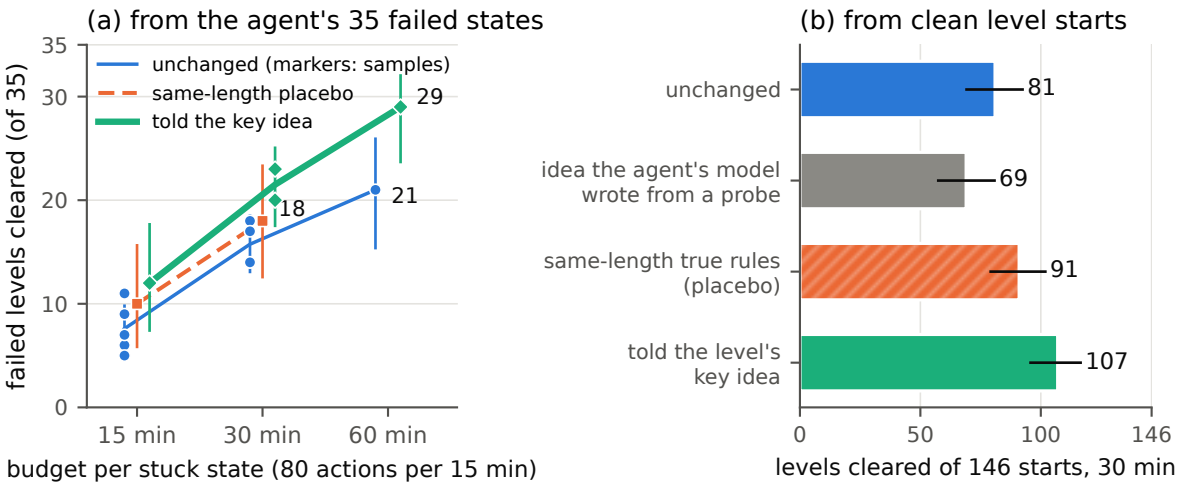


Figure 2: Levels cleared by budget. (a) From the 35 stuck states: every unchanged sample, the same-length placebo at two budgets and the told idea at three; a marker is one arm, a line joins the means. (b) From the 146 clean level starts that have a same-length placebo text, at 30 minutes: unchanged, the idea written by the agent's own model, the placebo (hatched) and the told idea, which clears 116 of all 157 starts. Interval bars are 95% Wilson intervals on the count, pooled over samples in (a). The two levers add.

Truncation does not measure time. A pre-registered within-run control counted the doubled-budget arm's clears made within the standard budget and expected them inside the floor; they fell below it (5 against 9 and 11; p 0.031 against the 11), because with a larger visible budget the agent spent actions more freely (Appendix D). A truncated long run is not a short run: for any agent that observes its own budget, time is read against independent samples at each budget, as every reading above was.

A supplied idea still needs the budget. Told the idea at the standard budget the agent cleared 12 of 35; at twice the budget 20 and 23; at four times 29. Time and the idea add rather than substitute. On the second checkpoint of the model, twice the wall time at the same 320 actions added nothing to the idea (22 against 24, p 0.625; 10 case-arms ended at the action cap against 2), so on that build moves bind and not time. And seven states that no unchanged or idea arm cleared at the standard or doubled budget were cleared, at four times the budget, by the unchanged agent in 3 cases and by the idea in 1; a paragraph locating on the board each object the idea names cleared 1 of 7. Where the agent's own route is longer than the budget, a stated idea does not shorten it.

5.2 Writing the idea

Can the agent supply the idea itself? Four arms ask, from four sources, and the answer is no in each.

Written by its own model. At each of the 157 level starts, before play, the agent's own model wrote the level's idea from what an agent can see: the start frame, the level number, the available actions and the one-step effect of each (greedy, thinking off, 256 new tokens). That text was then the first message, by the same path as the oracle's. It cleared 76 against the unchanged 90 (12 levels only with it, 26 only without, p 0.034), and on the 146 placebo levels 69 against the placebo's 91 (p 0.0003). The texts matched the oracle's key elements at 68 of 157 levels, and the agent thought less per move with them (1,782 tokens per action against 2,064) while clearing less: a message in the form of an idea does not help by its form, and a wrong one costs levels. This is one run of one writer; it rejects that writer, not self-written ideas in general.

Proposed at each failure. Asked, in a separate request at each failure, for three uses of the mechanics it had seen with a test of at most five actions, the same model's proposals entered the agent's next prompt. The arm cleared 17 of 35, inside the unchanged range. Coded against the key ideas, the proposals named the right mechanic without its key

use in 8 of 15 unsolved cases and the full key use in none of 24; 50 of 52 proposer calls were cut at their thinking budget.

Learned from other games. A 27B writer with an adapter tuned, per fold of five held-out games, on written key ideas of the other twenty cleared 13 against 19 for the same weights with the adapter off (p 0.18): no transfer. Against the oracle's text (23) the tuned writer is 2 against 12 discordant (p 0.013).

Drawn from a library. A library of 57 game-independent patterns abstracted from the other games' rules cleared 16 against a same-run unchanged 17. A pattern matching the level's idea existed for 16 of the 22 mechanics-only levels, but the harness put it in the first message in only 6 of 35 case-arms; in full public games the same patterns, with 308 further tricks, did not raise the score either, whether the model or the harness picked them; placed in the lookup message instead of the system prompt, the harness's picks read +5.15 (p 0.116), short of their rule's gain claim. A written protocol to keep a one-sentence rule, predict before each batch and rewrite the rule after a failure cleared 14; the agent followed it in a minority of its turns, and its last rule stated the key use in 6 of the 14 cleared cases and in none of the 21 uncleared.

An offline search for a signal that would let a scheduler keep the agent on levels it is close to clearing found none (AUC 0.53 for the harness's own priority; a perfect detector would gain about 1.1 points gross and lose once the displaced time is charged). The answer to RQ3: time raises clears and adds to the idea rather than replacing it; ideas from the agent's own sources do not raise clears, and the one its own model writes lowers them. The idea step cannot be bought with more of what the agent already does; Section 8 says what might buy it.

6 A world model is not the idea

The leading proposal for interactive benchmarks is a world model: learn the environment's dynamics, then plan (Ha and Schmidhuber, 2018; Hafner et al., 2023; LeCun, 2022). The agent of Section 4 already has most of the dynamics, the mechanics, in 22 of 35 failures. RQ4 asks whether a world model, given or learned, is enough; we give the agent one in four forms, from the perfect to the fitted.

6.1 Dynamics knowledge and goal knowledge

On the 22 mechanics-only levels at the doubled budget, a sandbox function that steps a copy of the real engine was put in the agent's hands, alone and with the idea (Table 3). Every case-arm called it, for about 5% of its wall time, and 15 of the 16 clears with both followed a simulated plan.

Table 3. Levels cleared of the 22 mechanics-only levels at 30 minutes, by what the agent was given (ARC-3).

	without the idea	**with the idea**
without a simulator	3, 7, 8, 8 (mean 6.5)	12, 12
with the real engine as a simulator	9	16

The idea adds about 5.5 clears without the simulator and 7 with it; the simulator adds about 2.5 without the idea and 4 with it (idea given the simulator: 16 against 9, p 0.039). A perfect model of the dynamics is worth more once the agent knows what to look for, and worth little before. Two of the four cells are single runs (the simulator alone, and with the idea); the arm is diagnostic, since a hidden game cannot be copied.

6.2 Planning with a perfect model

With the real engine as the model and the level's win as the goal, breadth-first search from each of the 183 level starts finds a plan on 18, 26 and 35 levels at 400, 2,000 and 10,000 nodes (30, 120 and 600 seconds); A* with a heuristic from the level's end frame on 13, 19 and 24; neither on any level whose shortest known solution exceeds 80 actions. Wall caps bind on most levels, so these are lower bounds for the node budgets, and still a perfect simulator gets bounded search to about one level in five. The plan that the idea gives is a plan the search does not find.

6.3 Models the agent writes

Offered, in a full public-game run, tools to write a step function for the level, verify it against the level's recorded transitions and plan with it, the agent called them on 25 of 481 levels, in 6 of 25 games, and scored 45.42 against 48.96 (p 0.27). Where verification returned ok, every level was cleared, but on 9 of those 11 levels the most transitions checked was 2 to 8: a model that agrees on two transitions has not been tested on the mechanic the level needs. With the tools enforced on the 35 states, so that after a failure the agent could not act until a step function had passed verification on the level (a verification gate), it cleared 16 of 35 and 8 of 22, inside the unchanged range, and 2 against 9 discordant with the told idea on the same base (p 0.065). The gate closed on 28 levels; the agent took a median of 149 seconds from its first failure to its first verification call, and 10 of 90 calls returned ok. Uptake, then verification, bind before planning does.

6.4 Models fitted to transitions

Replacing writing with fitting, a library of parameterised law kinds drawn from the other 24 games was fitted to each level's own transitions. Before the run, on the 35 states' frames at the cut, a single law matched at 0.9 or better on 16 of 35 levels and a composed model on none (median match 0.11); the law kinds fully express 107 of the 701 laws recorded for these games by human players (0.15), and none of the laws of 5 games. In the run the agent called the fit in 7 and 9 of 35 case-arms, and its own fits reached 0.9 on 4 levels where a silent fit by the host reached it on 12 and 16, so the library was not exercised and its rule gives no decision. Its clears (20 of 35) came late and none followed a plan built on it (7 executed, 0 completed). The second arm, which tested the model's proposals in the best fitted law before the agent saw them, cleared 10 of 35 in the same run, losing 10 cases the first arm cleared and gaining none (p 0.002), all among the 27 cases where the proposer fired.

The mechanism is instructive for anyone who scores a learned world model by next-step prediction. Of 103 simulated tests, 82 predicted that nothing would change, and 75 of those 82 came from models built on a gate law or a strip law: background mechanics whose checked transitions changed cells on a median of 16% and 2% of occasions. Because a transition in which nothing changes counts as a match, those laws scored 0.98 to 1.0 and 0.90 to 0.98 and ranked first, above the click and movement laws that the proposals actually tested (which change cells on 87% and 62% of their transitions). On grids, where most cells are still on most steps, a per-step match rate is a measure of stasis, and a model that predicts stasis will be the best model by that measure and the worst advisor. A score of the same fitted laws on the transitions each level's engine-verified rule names is planned for the next version (Appendix G).

6.5 Reading

The answer to RQ4: a world model is not sufficient, and it pays more after the idea than before. Given a perfect one, the agent clears few more levels until it is told the idea; bounded search over the perfect model finds plans for one level in five; models the agent writes are rarely built and rarely verified; models fitted to its observations are confident where nothing happens. The idea closes the gap that the model opens, not the other way round. For an agent that already has the mechanics, which is the agent on most of its failures, the next unit of effort belongs to the idea.

7 People, and the static case

RQ5 asks whether these levels are hard for people, and whether the gap holds beyond this model.

7.1 People clear these levels

For the 25 distinct (game, level) pairs among the 35 failed attempts, the public players who reached the level on the same game version cleared it at a median rate of 1.00 (range 0.50 to 1.00; 3 to 14 players per level), against a median of 0.92 over all 180 levels reached by at least three players. Several levels the agent failed were cleared by every player who reached them. The samples are small and players who reach late levels are self-selected, so this says that the levels are not unusually hard for people, not what a human clear rate on the hidden set would be.

A think-aloud dataset on the same games (Liu et al., 2026) marks, per trajectory and mechanism, the action at which a player first shows the mechanism. People took longer to recognise the key mechanism at the levels the agent clears only when told the idea than at the levels it clears unchanged: a median of 58 actions against 23.5 (difference 34.5, interval 6.25 to 55.0, under a rule fixed before the outcome). At the 25 stuck levels, 74 of the 145 human recognitions of the key mechanism came at an earlier level: the idea at a stuck level is often a new use of a mechanism people had already recognised, which is the gap this paper finds for the agent. The readings are descriptive: of 107 intervals computed, 14 exclude zero against about 5.4 expected by chance.

7.2 The static case: ARC-AGI-2

ARC-AGI-2 asks for the output grid of a transformation shown by a few demonstration pairs. There the idea is the transformation and a checker exists in the demonstrations themselves. We audited a public 4B test-time-training transducer of the NVARC family (Sorokin and Puget, 2025), the base of our entry, on 40 evaluation tasks with 65 test outputs: it was right on 12 of 65, and its second attempt never added a solve. The right grid appeared anywhere in its candidate pool for 15 of 65, so a perfect selector could add at most 3; for 50 of 65 it was never generated. Two identical runs were right on 12 and 16, and a wider-search variant on 14; 12 outputs were right in all three runs and 16 in any, against 33.7 expected if the runs drew independently at their rates, and 49 in none. Generation, not selection, is the limit, and the limit is a property of the task, as on ARC-AGI-3.

A 27B model of the same family writing programs and checking them on the demonstration pairs, with the checker as a perfect verifier, solved 3 of 65, all of them tasks the 4B already solved, and exhausted its token budget on 38 of 40 tasks. The same model answering grids directly was right on 9 of 65, 7 of them outputs the 4B missed, so the two are complementary (union 23 against 16); on 20 unseen tasks chosen by length before the run, the 4B was right on 9 of 25 outputs, the 27B on 14 and their merge on 16. The entry routes the shortest tasks to the 27B after the 4B. On a 145-output bank the 4B alone is right on 40, the merge on 45 and the merged pool's oracle on 55; more 27B samples raised the oracle to 61 of 139 and lowered the merge to 43, because wrong 27B grids replaced right second attempts. On the leaderboard the entry read 34.31 and, resubmitted byte for byte, 29.31.

The shape is the one of Section 4. The answer to RQ5, so far: the levels are not hard for people; the told idea raises clears on a second checkpoint of the same model (Section 4.2); in the static case a perfect checker, like a perfect simulator, does not substitute for generating the idea, what is solved is a stable set, and one reading is not a result; and whether other model families show the gap is open (Section 9).

8 Discussion: what to build, and beyond ARC

8.1 Three open problems, answered for one agent

The benchmark's hosts name three open problems: how to separate knowledge from reasoning, why the efficiency gap is bottlenecked by ideas rather than engineering, and how benchmark design must adapt to overfitting. Knowledge and reasoning come apart as the mechanics and the idea, and for this agent the first is mostly present and the second mostly absent. The efficiency gap has a size: a supplied idea raises clears from 15.75 to 21.5 of 35 at the same budget and the full-game score from 48.96 to 66.06, and still needs about twice the budget to act on; the budget spent without the idea is the inefficiency, and more of the same spending does not buy it. And replayed failure states with a floor, a placebo and a ceiling cannot be overfitted the way a score can: an agent tuned to the public games is measured on what it does from its own failures, against its own repeated runs.

8.2 A build order

The answers to RQ1 to RQ4 give an agent of this kind a build order.

1. **Per-game token budget before anything else.** On stuck levels, clears go from 5 to 11 at 15 minutes to 21 at 60. The budget is bought by throughput, not by concurrency on the same hardware: raising concurrent streams from 10 to 12 cut tokens per game-pass from 175k to 155k and the score from 48.96 to 42.87 (p 0.068). The best-ranked public ARC-3 notebook after the close, by its code, differs from this one mainly in serving many more

games at once with a pruned model and parked sessions, which is throughput engineering; whether that is what moved its score is not measured here.

2. **Then the idea, found rather than told.** The 22 mechanics-only levels are the test set, with per-arm clear rates known (Table 1); the told idea is the ceiling (16 of 22 with a simulator, 17 of 22 at four times the budget) and the proposer is the baseline (right mechanic in 8 of 15, key use in none). Whatever finds the idea is scored by how many of the 22 it converts against those two numbers, at a stated budget, with a placebo arm. The arms of Section 4 measure what a told idea is worth on fixed stuck states; the leading entry relied on that worth, through its reviewer and its learning from stuck states, without measuring it (Chen, 2026).
3. **A world model after the idea, not before.** It pays once the agent knows what to look for (Section 6). Verification before planning, and a stasis-corrected score for any model fitted to transitions.
4. **Not these.** Context handling beyond what the harness does, guard repairs, prompting, reasoning effort, cross-game libraries, a learned writer, scheduling by progress signals, and a larger model of the same family in the same harness.

8.3 Efficiency

Table 4. Tokens per action and clears by arm (ARC-3; from the run records; tokens per action are over the level set in the clears column, so the 157-level rows differ slightly from the 146-level figures of Section 4.3).

Arm	Budget	Tokens per action	Clears	Note
unchanged, level starts	30 min	2,064	90 of 157	
told the idea, level starts	30 min	1,601	116 of 157	thinks less per move, moves more
placebo, level starts	30 min	1,996	91 of 146	
self-written idea, level starts	30 min	1,782	76 of 157	
second checkpoint, unchanged	30 min, 320 actions	482	17 of 35	cheaper moves, not faster: prefill binds at about 83k tokens per request
unchanged, full games	per-game budget	836	48.96	
told the idea, full games	per-game budget	775	66.06	54 game-passes at the wall against 71

Generated tokens per clear on the 35 states (from the run records; completion tokens summed over the 35 case-arms, divided by clears): told the idea at 30 minutes 1.98M over 23 clears, 86k per clear; unchanged at 30 minutes 2.46M over 18, 137k; told at 60 minutes 2.90M over 29, 100k; unchanged at 60 minutes 4.37M over 21, 208k. The idea cuts the generation spent per cleared level by about a third at 30 minutes and by half at 60, and the agent that has it stops sooner: about 1,288 model calls (turns plus read-backs) at 60 minutes against 1,828 unchanged.

Cheaper moves alone bring little: at move costs as low as the told-idea arms', the second checkpoint cleared 17 against their 21.5, and the ceiling on what cheaper moves buy is about 1.25 clears. What the idea buys is not fewer tokens; it is a shorter route.

8.4 Beyond ARC

Three statements travel.

- **One run is not a result.** A game's score has a median standard deviation of 16 points over four passes of one notebook; a run mean of four passes has a standard deviation of about 2.2 to 2.7; two identical submissions read 34.31 and 29.31. A single whole-task run reports noise of that size. Paired continuations from replayed states, with repeated unchanged samples, read effects a tenth of it. Entrants report the same (Section 2): a post-competition thread holds that the competition rewarded inference throughput and that with one scored run a day a harness change was hard to tell from luck (Makarov, 2026); the sixth entry's 34 harness variants read 0.96 to

7.22, and its own stand of one prompt change on 36 saved stuck states, with two identical arms, read 9, 9 and 11 new levels (Zatravkin, 2026).

- **A correct idea is usable before it is generable.** Told the idea, the agent clears 116 against 90; writing the idea itself, 76 against 90. The gap between what a model can use and what it can produce is 40 levels of 157 here, the quantity that oversight proposals which supply guidance rely on; the measure gives it a number per agent.
- **Guidance needs a placebo arm.** A same-length block of true but irrelevant rules recovered 38% of the idea's gain from level starts and, on the stuck states, between about 40% and 70% depending on the budget and the comparator. A claim that a hint, a memory or a retrieved document helped an agent is a claim about content only if a text of the same shape and length did not help as much.

9 Conclusion

The five questions, in order. RQ1: on its failures the agent has the mechanics and lacks the idea. RQ2: told the idea it clears more, from stuck states, from clean starts and in full games, and a same-length placebo recovers part of that gain; from level starts the idea's margin over the placebo meets its pre-set size and misses its pre-set p by 0.001, a reading of report. RQ3: time raises clears and adds to the idea; the agent cannot yet write the idea, and given it still needs about twice the budget to act on it. RQ4: a world model is not sufficient, and pays more after the idea than before. RQ5: the levels are not hard for people, the told idea raises clears on the same model's other checkpoint, the static case shows the pattern, and other model families are open. An open LLM agent on tasks it has never seen therefore fails at the idea, not at memory, tools, time alone, a world model or model size. The idea gap of Section 3 measures this for any agent on a public fixture, and the readings for the best open agent put the next unit of effort on finding the idea, after the budget to act on it and before a world model.

Limitations

One model family, in one harness, on one benchmark's public games, whose rules and solutions exist in a form hidden games do not have; a second checkpoint and a static benchmark show the same shape, and a second family and frontier models, the open part of RQ5, are the first thing to add. Most arms are single runs of 35 cases; the design detects large effects, and "inside the unchanged range" bounds an effect rather than excluding it. The told idea's margin over a same-length placebo passes its pre-set size and sits just above its pre-set p from level starts (p 0.011) and is not established on the stuck states; the full-game and 60-minute arms have no placebo. The rule-discovery labels come from one labeller who was not blind to the continuation outcomes; The self-written arm rejects one writer, not self-written ideas. Where a told idea should sit in the context is open: the sixth entry's attention logs show its model reading board images at less than half, and instructions at less than a third, of their share of the prompt (Zatravkin, 2026); the arms here do not decide placement. The human comparisons are descriptive, on small and self-selected samples.

Ethical considerations

Human participants. The team's own human data is its play: members of the team played the public games individually, by hand; their texts serve only as inputs to arms, and no timings or personal data are used. Two public datasets of human play are used for analysis only and are not redistributed: one carries no licence statement and is reduced to per-level counts; the think-aloud dataset is CC BY 4.0 and is reduced to recognition counts (Appendix H). The competition's human baseline action counts are never read by any policy reported here.

Data and licences. Released under the collective's name: the rules for 25 games with their engine checks and the probe tool; the 35 case bundles with replay and restore; the 22-level test set and the 157 level starts with per-arm clear rates; the analysis code and every decision rule as it was frozen (code MIT-0; rules, cases, readings and the test set CC BY 4.0, with attribution to ARC Prize for the MIT engine). The rules quote short snippets of the public game source with line numbers; the game files themselves, the harness, model weights and third-party packages are not redistributed, and competition imagery is not reproduced (Appendix E).

Compute. The ARC-AGI-3 runs reported in the body each ran on one GPU: public-game runs of about 8.2 GPU-hours each, and 35-case continuation runs of about two to four hours (Appendix D); the ARC-AGI-2 entries ran on four L4 GPUs (Appendix B). No proprietary model was called in any experiment reported in the body.

Use. The measure is an evaluation method for agents on public games; it releases no model, no capability and no personal data.

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Appendix A. The arms, as built

Every arm continues from the same 35 replayed states with the restored conversation and functions of Section 3.2, except A3 (no conversation, no functions), A4 (conversation trimmed) and A5 (no functions). Budgets: standard 80 actions or 15 minutes; doubled 160 or 30; quadrupled 320 or 60. Flags off reproduce the unchanged harness exactly (unit-tested).

Table A1. Arm definitions.

Arm	Change from the unchanged harness	Budget
A0, A0b, A0c, A0d, A0e	none; five independent samples	standard
A1 probe-first	a notice asking for two hypotheses and a probe of at most 6 actions before long plans	standard
A2 timeout repair	tool timeout 120 s instead of 30 s; queued actions cancelled on timeout; functions kept	standard
A3 fresh start	no conversation, no retained functions; one line stating the level and actions so far	standard
A4 trim	conversation cut to its most recent 45% of messages, as one of P0's own trims does	standard
A5 no functions	full conversation; no retained functions	standard
A6 repertoire sweep	every 25 actions without progress, the observed effects per action are listed and the agent asked to try known mechanics in new roles	standard
A8l low effort	reasoning effort low instead of the template's default, the highest	standard
A7, A7w, A7c	the level's key idea (A7), its win condition only (A7w), or a length-matched block of true but irrelevant rules (A7c), in a message at the restore	standard
A11q restart option	the game's RESET action exposed to the agent	standard
A0L, A0Lb (twice), A0Lc	none; four independent samples	doubled
A7L told the idea	the key idea in a message after the restore; run twice	doubled
A7cL placebo	A7c's text, byte for byte	doubled
A9 trick library	57 game-independent patterns from the other games' rules, ranked by observed effects	doubled
A7G, A7B learned writer	the idea written by a 27B tuned per fold on other games' ideas (A7G) or untuned (A7B)	doubled
A21 proposer	at each failure the same model proposes three uses of the mechanics seen, each with a test of at most 5 actions	doubled

Arm	Change from the unchanged harness	Budget
A10l, A10x, A10m 27B agent	the 27B of the same family as the agent's model, at low, full or medium effort	doubled
A11r (twice), A12n, A13	restart option; with a prompt line to restart after no progress (A12n); with the told idea (A13)	doubled
A12S, A12SI simulator	a sandbox function stepping a copy of the real engine, alone or with the idea; 22 levels	doubled
A20 rule card	one message: keep a one-sentence rule, predict before each batch, rewrite after a failure	doubled
WM2 tools	write a step function, verify it on recorded transitions, plan with it; actions refused after a failure until verified	doubled
A22L, A22P law library	parameterised law kinds fitted to the level's transitions, as tools; A22P adds the proposer with each test run in the best fitted law	doubled
A15, A16 second checkpoint	the ThinkingCap checkpoint of the model, unchanged (A15) or with the idea (A16)	30 min, 320 actions
TCI60	A16 with 3,600 s per case, actions kept at 320	60 min, 320 actions
A0H, A11h (twice), A14	unchanged; restart option; told the idea	quadrupled
A8L, A7Lr, A0X, A7X	the residual seven only: the idea plus a paragraph locating its objects; the idea repeated; unchanged and the idea at four times the budget	doubled; quadrupled
A0F, A7F level starts	157 public levels with an engine-checked idea, from a hash-checked level start, unchanged or told the idea first	doubled
A7cF placebo from starts	the same-length control text as the first message; 146 levels	doubled
A7SF self-written from starts	the agent's own model writes the idea from the start frame, the actions and their one-step effects; that text first; 157 levels	doubled

Table A2. The standard-budget arms of the first four runs, paired with the first unchanged sample (ARC-3).

Arm	Cleared of 35	Discordant, arm only / unchanged only	McNemar p
A0 unchanged	9		
A0b unchanged, second sample	11	4 / 2	0.69
A1 probe-first	5	4 / 8	0.39
A2 timeout repair	6	3 / 6	0.51
A3 fresh start	1	0 / 8	0.008
A4 trim to 45%	7	2 / 4	0.69
A5 no retained functions	6	2 / 5	0.45
A0L, doubled budget	14	7 / 2	0.18

The two unchanged samples agree on 29 of 35 levels (7 cleared by both, 22 by neither, 6 discordant). Across the thirteen standard-budget arms of the first six runs, 23 of 35 levels were cleared by at least one arm against 33.3 expected under independent draws at the observed rates, and 12 by none (permutation p 0.019).

Table A3. Clean level starts on the public levels with an engine-verified idea (30 minutes, 160 actions; paired by level; ARC-3).

Levels or measure	unchanged	self-written idea	same-length placebo	told the idea
cleared, all 157	90	76		116
cleared, the 146 with a placebo text	81	69	91	107
cleared, shortest solution 1-10 actions	21			22
cleared, 11-20 actions	37			44
cleared, 21-40 actions	21			30

Levels or measure	unchanged	self-written idea	same-length placebo	told the idea
cleared, 41-80 actions	11			20
cleared, 81 or more actions	0			0
cleared within 80 actions and 900 s	56			80
tokens per action (157)	2,064	1,782		1,601
tokens per action (146)	2,078		1,996	1,615
actions per case-minute (146)	1.99		2.06	2.84
case-arms ending at the 30-minute wall (146)	65		52	38

Pre-set rules: told the idea generalises if the gain is at least 10 with p at most 0.01 (met: 26, discordant 35 against 9, p 0.0001); content supported over the placebo if the margin is at least 10 with p at most 0.01 (read: report, 16, discordant 26 against 10, p 0.011); self-written helps if the gain is at least 10 with p at most 0.01 (read: no help, −14, discordant 12 against 26, p 0.034).

Appendix B. Leaderboard entries, both tracks

The competition scored one submission per team per day on the hidden set; two were selected as final per track. All readings are from the submissions ledger. ARC-3 entries ran on one RTX PRO 6000; ARC-2 entries on 4×L4.

Table B1. ARC-3 entries.

Date (UTC)	Submission	Score	What it was
Oct 17	57310234	27.01	P0, the public harness notebook unmodified (its author reports 27.89)
Oct 18	57337312	25.62	P0 again, same notebook version: the submission-to-submission spread
Oct 19	57387077	27.94	P0, third reading (mean of three 26.86, SD 1.17)
Oct 22	57519241	23.10	P0 with a public slot-priority formula and a start-up keep-alive
Oct 23	57541952	0.00	P0 with the restart option exposed, no start-up protection; a submission that sends no action within 15 minutes scores 0, and the model load that day took 25 to 30 minutes
Oct 24	57581028	21.32	P0 with the restart option, keep-alive and a 2-hour model-load wait
Oct 25	57610430	24.03	P0 with keep-alive and the load wait (A1)
Oct 26, 27	57646234, 57682209	25.92, 22.42	P0 with the slot-priority formula, keep-alive and the load wait (A2), twice
Oct 28	57716633	27.54	A1 on an FP8 build of the same model (124.8 GB to load against 181.2)
Oct 29	57752194	16.53	A1 with the ThinkingCap checkpoint of the model
Oct 30	57785925	30.26	the FP8 build again, same kernel version: **the primary submission of this paper** (`drdiwakarmishra/arc-agi-3-p0-ka2-fp8ple`)
Oct 31	57833374	23.67	the FP8 build with a 365-entry pool of patterns and tricks in the system prompt, picked by the harness from cue tags (HLB)
Nov 1	57862352	26.51	HLB's picks at the start of the lookup turn's message, system prompt unchanged (HLC)
Nov 2	57887330	29.13	the FP8 build with up to 15 of the 57 patterns placed once in the system prompt at the first call (HLS; `drdiwakarmishra/arc-agi-3-fp8-hls`)
Final selections	57785925, 57887330	30.26, 29.13	the FP8 build and HLS

On the public games P0 scores 48.96 (25 games, 4 passes; pass means 51.4, 52.1, 45.2, 47.1; 108.0 levels per pass; 29 of 100 game-passes won), roughly twice its leaderboard reading; other teams report similar or larger public-to-hidden drops. The public-game variants are in Table B3.

Table B2. ARC-2 entries.

Date (UTC)	Submission	Score	What it was
Oct 17	57295060	0.00	a CPU program-search entry, as its local evaluation predicted
Oct 19	57387075	30.56	B0, the public 4B test-time-training notebook unmodified
Oct 22	57518886	28.89	B0 with half the test-time training (H1)
Oct 23	57541839	31.39	size-routed hybrid v1: H1, then the 27B answering directly on the shortest tasks in the time left
Oct 24	57582551	31.81	hybrid v2
Oct 25	57610305	28.61	v6nb: quarter training, half-view search, deadline caps
Oct 26	57646090	34.31	**v7-vr**: v7 with verify and refine (the 27B's program run on the demonstrations, one repair, one rescue); final selection (`purushottamanurag/arc-agi-2-hybrid-v7vr`)
Oct 27	57682082	30.97	v7: v2's B0, then the 27B direct with deadline caps
Oct 28, 29	57716508, 57752087	32.64, 32.22	v7-vr2: written grids ranked above outputs of unverified programs, twice
Oct 30	57785824	29.31	v7-vr again, byte-identical: the 5.00 gap is run-to-run variation
Oct 31	57837368	31.81	S1: v7-vr2 with 128 training sequences
Nov 1	57853811	32.64	zbeam: v7-vr2 with an attempt-2 picker from B0's own signals; final selection (`purushottamanurag/arc-agi-2-hybrid-v7vr2-zbeams`)
Nov 2	57887325	29.31	amax: v7-vr2 with another attempt-2 picker (scored after the close)
Final selections	57646090, 57853811	34.31, 32.64	v7-vr and zbeam

A difference of two hybrid readings has a standard deviation of about 3.2 (2.6 for copies of B0 alone); none of the ARC-2 differences above is a measured gain.

Table B3. ARC-3 public-game runs (25 games × 4 passes, same hardware), against P0.

Run	Change from P0	Mean	Δ vs P0	p (perm.)
P0	none	48.96		
I3	a computed mechanics summary as a sandbox variable	46.83	−2.12	0.55
I3b	I3 plus a 534-character prompt section	43.89	−5.07	0.12
I5	other teams' solved-level memory and strategy audit	45.69	−3.27	0.35
I6	scheduler stall falloff and tail fade changed	46.87	−2.09	0.53
I7	12 concurrent streams instead of 10	42.87	−6.08	0.068
RS, RSb	restart option exposed, keep-alive; two runs	44.67, 47.00	−4.29, −1.96	0.21, 0.56
WM1	optional world-model tools	45.42	−3.54	0.27
FPG	the FP8 build of the same model, no keep-alive	44.21	−4.75	0.163
TCF	the ThinkingCap checkpoint; 25 games × 2 passes	25.71	−23.25	< 0.0001
HLA, HLB	FPG with the 365-entry pool in the system prompt, picked by the model (HLA) or by the harness (HLB)	36.79, 41.27	−7.42, −2.94 vs FPG	0.029, 0.376
HLS, HLC	FPG with up to 15 of the 57 patterns once in the system prompt (HLS); HLB's picks in the lookup message (HLC)	42.29, 49.35	−1.91, +5.15 vs FPG	0.55, 0.116
OIF	the level's key idea shown once at the first call in each of 157 levels	66.06	+17.11	< 0.0001

No deployable variant improved on P0, and with four passes per arm effects smaller than about 10 points could not have been detected. HLB's loss was throughput: its hint section was replaced at each lookup, the prefix-cache hit fell from 93.6% to 86.0%, and it took 17,190 actions against FPG's 20,143; at equal actions it was level with FPG.

Appendix C. Ground truth and labels

Rules from the game source. For each of the 25 public games the rules were written from the game's source (its step logic, win and lose conditions and level list): what each action does, the core rules, and per level the layout, what is new, the key insight and a winning strategy. Each statement ends with its source line range, a claim id and a status. Where a statement has a mechanical consequence it is checked on the engine: the check reaches the level by replaying a human winning session's earlier levels or by jumping to it, runs an action sequence, and judges a recorded expectation (cleared on the last action and not before; not cleared; game over; no game over). Statements without a mechanical expectation are marked inspect. A probe tool re-runs any recorded check with per-step board diffs. The rules were written before the failure dossiers were read, and no key insight was changed afterwards. Counts: 794 claims, 629 engine-checked and 165 source-only; re-running every check gives 547 pass, 82 inspect and 0 failures; on the 25 failed (game, level) pairs, 101 claims (85 pass, 5 inspect, 11 source-only), and every failed level's key insight and win condition is engine-checked, usually with a contrast showing that the naive alternative does not clear.

Failure dossiers and categories. For each of 43 sampled cases (the 35 failures, plus 8 cleared levels matched to failed ones) a dossier of at most 25 KB summarises the budget, every action with its board change, and every model turn with a short quote. Two labellers labelled the dossiers independently, the second blind to the first, with a primary failure category (perception, exploration, modeling, goal inference, planning-execution, context-transfer, harness-runtime, uncertain), a confidence, an alternative, and fields for the first incorrect belief, prediction mismatches, repeated actions, guard and timeout events, context trims, and whether a better next experiment was visibly available. They agreed on the category in 23 of 35 (kappa 0.54) and in 32 of 35 when either's alternative counts; the consensus cells are modeling 9, harness-runtime 5, planning-execution 4, goal inference 4, exploration 1.

Rule-discovery labels. Against the engine-checked key insight of each level: yes if the agent stated it, or an equivalent, as its working belief at some turn on the level; partial if its core mechanics were right and the insight missing or half-stated; no if the mechanics were wrong; unclear if the dossier is too thin. Every evidence quote is checked verbatim against the dossier by script. Labels for the 35 failures: yes 11, partial 22, no 2, unclear 0. One labeller, not blind to the continuation outcomes; the labels are drawn only from P0's own dossiers.

Appendix D. Pre-registration, statistics and determinism

Rules fixed before runs. Every continuation run after the first had its arms and decision rule written down before it ran, from the fifth on as a committed rule file; the later runs have rule files that name the thresholds quoted in the text (for example, the level-start run generalises at a gain of 10 with p at most 0.01; its placebo arm supports content at a margin of 10 with p at most 0.01; its self-written arm helps at a gain of 10 with p at most 0.01; the placebo at the doubled budget supports content at 17 or fewer, rejects it at 20 or more and reads report between; the enforced tools promote at 22 of 35 or 11 of 22; arms on a second model family read the same shape at a gap of 6 or more with p at most 0.05 and at most 25 unchanged clears, no gap at a gap of 2 or less, and closed by capability at 26 or more unchanged clears; the other thresholds are listed in Section 3.6). A reading that meets its rule is stated as met; one that does not is stated as not met or report; no threshold was changed after a result.

Continuations. Cut rule: half-way through the actions spent on the failed level (the level start if fewer than 4), snapped back to the last turn boundary; the snap is 0 actions in 9 cases, at most 6 in 28, and 8 to 33 in the remaining 7. Outcome: the failed level cleared or the game won. Per arm against the unchanged arm: cleared counts, discordant pairs (b, c), exact two-sided McNemar $p = \min(1, 2 \cdot P(X \le \min(b, c)))$, $X \sim \mathrm{Bin}(b + c, 0.5)$; the same against each further unchanged sample. Union over arms against the expectation under independent draws at each arm's rate. Permutation test of "arms do not matter": statistic $\Sigma\, n_a\, (p_a - \bar{p})^2$, arm labels shuffled within case, 20,000 draws. Power by exact enumeration: an arm that clears a superset of the unchanged levels is detected at 80% power if it adds about 22 percentage points; with 10% reverse discordance it must add about 36. Pooled comparisons across runs are unpaired and indicative.

Whole-game runs. 25 public games × 4 passes in one session with each game's time share matched to a hidden run; scores recomputed with the official formula match all 600 recorded game-passes of six runs. Variant comparison: mean over games of the per-game difference of pass means; game-paired permutation test at pass level, 20,000 draws; game-level percentile bootstrap as a secondary interval. Noise: pooled within-run SD of pass means 4.91; observed SD of six run means 2.21; per game, a median SD of 16.4 over P0's four passes; 80% power at a difference of about 10 points. Passes within one session are not independent replications, so pass-level p-values are exploratory.

The internal control that failed. A pre-registered within-run test counted the doubled-budget arm's clears made within the standard budget and expected them inside the floor; they fell below it (5 against 9 and 11; p 0.031 against the 11), because with a larger visible budget the agent spent actions more freely: four of the nine levels it had cleared by 15 minutes used more than 80 actions, and three it cleared within 80 actions took longer than 15 minutes. Time is therefore read only against independent samples at each budget.

Determinism. All 35 prefixes replay board by board against the recorded events locally (35 of 35) and on the competition's own wheels in a CPU preflight (35 of 35, 8,558 actions); replay through the harness session path reproduces runtime state, history and events (7 of 7 sampled cases); every GPU case-arm of the first four runs ran with no replay mismatch (280 of 280); deferred state writes during replay give byte-identical files (unit test); flags off reproduce the unchanged harness (unit test); 130 of 144 human winning sessions reproduce the win on the competition engine, 114 frame by frame; the bundled scorer and the local one have identical calculators and reproduce all 600 recorded game scores.

Compute. Public-game runs about 8.2 GPU-hours each; the first four continuation runs 3 h 10 min, 117, 116 and 108 minutes; later 35-case runs about 2 to 4 hours each on one RTX PRO 6000.

Appendix E. The kit, assets and licences

Released under the collective's name (code MIT-0; rules, cases, readings and the test set CC BY 4.0, with attribution to ARC Prize for the MIT engine): the rules for 25 games with their engine checks and the probe tool; the 35 case bundles with the replay and restore code; the 22-level test set and the 157 level starts with per-arm clear rates; the analysis code; every decision rule as it was frozen; the derived human counts. Not redistributed: the harness, model weights, third-party wheels, competition game files and imagery, and the human-play datasets, which are cited.

Table E1. Assets used.

Asset	Use	Licence or status
the public harness notebook (D. Franzen's Milestone 2 notebook, derived from J. Cottaar's Duck notebook and the Tufa Labs Duck harness)	P0, unmodified; our public copy replaces the introductory cell with a credit note	Apache-2.0 (notebook); the Duck harness states MIT in package metadata, no licence file in its repository
Qwen3.8-Flash-Next; the W4A16 AutoRound quantisation with an MTP draft head; the FP8 build of the entry	the agent's model	Qwen Community License 1.0
the ThinkingCap checkpoint of the same model (public Kaggle model files)	arms A15, A16, TCI60; the TCF run and one entry	Qwen Community License 1.0 (the ThinkingCap model card and its W4A16 build's card, read 2026-11-09); the Kaggle datasets that mirror the shards state "Other (specified in description)" and defer to the original models' licences; the Kaggle model page states Apache 2.0
Qwen3.8-27B-FP8	arms A10l, A10x, A10m; the ARC-2 hybrids	Apache-2.0 (model card, read 2026-11-09)
world-model tools for the sandbox (step function, verification, planner), with our labelled patch	WM1, WM2, R1z	the team's own code (its harness package, MIT-0), applied as a patch to the Apache-2.0 notebook code; nothing third-party

Asset	Use	Licence or status
the SGLang fork wheelhouse used for serving	P0	Kaggle licence field unknown; upstream Apache-2.0
arc-agi 0.9.9 and 0.9.8, arcengine 0.9.3	engine and official scorer	MIT
ARC-AGI-3 public game files	replay, rules extraction	competition data; not redistributed; rules quote short snippets with line numbers
human-play dataset (Hugging Face, 340 sessions)	analysis only; per-level counts published	no licence stated; unofficial uploader
think-aloud dataset (Liu et al., 2026)	recognition counts; derived numbers only	CC BY 4.0
the team's own play, each game played individually by hand	inputs to arms A20, LI1, HLA, HLB, HLC and R1z; solution lengths	the team's own; the hint texts are public (CC0)
the ARC-2 base notebook and its model	the ARC-2 entries	Apache-2.0 (Kaggle pages); training-data licences not verified
the public slot-priority cell, other teams' memory and audit mechanisms	one ARC-3 entry; variants I5, I5a-16	Apache-2.0; re-implemented and credited

The human baseline action counts the competition exposes during submission are never read by any policy we wrote. No proprietary model was called in any experiment reported in the body of this paper. Competition game imagery is not reproduced.

Appendix F. ARC-AGI-2 in full

The base and the audit. B0 is the public 4B test-time-training notebook unmodified (30.56 on its first reading). On 40 evaluation tasks with 65 test outputs, scored against the competition's solutions: pass@2 12 of 65 and pass@1 also 12, so the second attempt never adds a solve; the right grid anywhere in the candidate pool for 15 of 65 (a perfect selector adds at most 3, two of them in one task); never generated for 50 of 65; every task stopped when its search ran out of sequences above a probability floor, not on time. An identical repeat scored 16; a wider-search variant 14; 12 outputs were right in all three runs and 16 in any, against 33.7 expected under independent draws, and 49 in none. Half the test-time training kept 15 of 65 in two runs at 18 to 20% less time per task; one epoch of adaptation on 80 evaluation tasks gave 15 of 65 (rule: adopt at 19; not met).

A second solver. The 27B as a program-writing agent that checks its programs on the demonstration pairs solved 3 of 65, all also B0's, and used up its token budget on 38 of 40 tasks (357 generated tokens per second). Answering grids directly it reached pass@2 9 of 65, 7 of them outputs B0 missed in both runs (union 23 against 16); its first sample is right on 17 of 44 outputs from the shortest quarter of tasks by length and on 2 of 46 beyond. The tasks it solved sat at B0-confidence ranks 9 to 31 of 40, so routing by B0's confidence does not reach them; routing by size does. On 20 unseen tasks fixed by length before the run (25 outputs): B0 with half training 9 of 25, the 27B 14, merged 16 (9 gained, 2 lost).

The entry. Phase 1: B0 with 64 training sequences per task and all 16 search views, starting no new task after 9 h 18 min. Phase 2: the 27B (FP8 weights and KV cache, two-token draft, 81,920-token context, highest reasoning effort), tasks shortest first, requests until 11 h 15 min, hard stop at 11 h 30 min. Verify and refine: each reply carries a grid and a Python transform; the transform runs on the demonstration pairs; one repair round follows a failure and one rescue request follows a cut. Merge: a program that reproduces every demonstration goes first; the 27B's answer replaces B0's attempt 2 when it differs. Readings: Table B2.

Measuring changes offline. An output bank of 101 unsealed public evaluation tasks and 145 outputs holds every B0 candidate pool and 27B sample; 19 further tasks are sealed for one final check. A rule is pre-registered, then replayed on CPU against the saved pools. Gate: net 3 or more outputs gained with at most 1 lost; in 800 level-preserving random replays no seed reached net +3. Calibration over four entries: board ≈ 8.84 + 0.727 × bank, residual SD 0.42.

B0 alone 40 of 145; the entry's merge 45; the merged pool's oracle 55 (B0's pool alone 46). From Oct 31 each entry carried a prediction committed before submission.

What did not pass (right outputs of 145 unless noted): attempt-2 pickers from B0's own signals (zbeam 40 to 41 alone, 45 to 47 merged; amax unchanged; neither net +3; the right grid is in B0's pool for 46 of 145, so selection has little left); colour 0 fixed in every permutation (40 to 39); re-ranking B0's pool by a 30B model's likelihood (35) or a 2B decision model as judge (at most 36); more 27B samples (pool ceiling 55 to 61 of 139, the merge 45 to 43: beyond the short band, wrong 27B grids replace right second attempts); worked programs in the rescue prompt (net −3 over 196 paired draws); 128 training sequences (36 against 40; board 31.81); another 30B model in the 27B's place (8 of 25 against 14 on the routing tasks); the agent's own model on 4×L4 (99 tokens per second, 0.30 of the 27B's, 4 of 25 right).

Limitations. All readings are on the public half, mostly one each; the bank's tasks are public and may be in the 27B's training data; the 27B takes one sample at its default temperature.

Appendix G. The world-model arms: details not in Section 6

Planning with the real engine. Breadth-first search from each of the 183 level starts with the real engine as the step function and the level's win as the goal finds plans on 18, 26 and 35 levels at 400, 2,000 and 10,000 nodes (30, 120 and 600 seconds); A* with a heuristic from the level's end frame on 13, 19 and 24; none on a level whose shortest known solution exceeds 80 actions (41 to 80 actions: 0, 0 and 4). Wall caps bind on most levels (98 timeouts at 10,000 nodes), so the found rates are lower bounds for the node budgets.

Tools, optional (WM1). Called on 25 of 481 levels, in 11 of 100 game-passes and 6 of 25 games. Verification returned ok on 11 levels, all cleared, but on 9 of them the most transitions checked was 2 to 8 (0 and 21 on the other two); never ok on 7 levels, 5 of them cleared anyway; a plan executed on 16 levels, all cleared.

Tools, enforced (WM2). Verification ok only with every available action covered; 60 s tool limit. Paired against the unchanged samples p 0.69 to 1.00; 2 against 9 discordant with the told idea on the same base (p 0.065). The gate closed on 28 levels; median 149 s from the first failure to the first verification call; 10 of 90 verification calls ok, a lower bound since every replayed transition must match.

Laws fitted to transitions (R1z). Fit at most 4.2 s per level; composed-model match median 0.11, mean 0.25, 7 levels at 0.5 or better. Of the 701 laws recorded for these games by human players, 177 (0.25) are fully expressible with the kind set from all 25 games and 107 (0.15) leaving each game out; 5 games have none. The library as tools: 4 clears within 80 actions and 15 minutes against 5 to 6 for the unchanged samples; 7 library plans executed, 0 completed; paired p 0.07 to 0.77. The proposals arm: 59 proposer calls, none cut at the thinking budget, 21.7 s each, 3% of wall time; in the 8 cases without a proposer call both arms cleared 7. Click and movement laws (87% and 62% of their checked transitions changing cells) gave 6 of 11 and 1 of 4 no-change predictions; the gate and strip laws (16% and 2%) 75 of 75. That the notice steered the agent away from the tests it needed is an inference not tested; one run cannot separate the notice from the medium-effort proposer. A stasis-corrected score of the same laws on the transitions each level's rule names is planned for the next version.

Appendix H. Human data

Public play. A public dataset of 340 human sessions on the public games (one row per session; created June 2026; its card attributes the data to ARC Prize, its uploader is not an official account, and no licence is stated). All 25 competition game versions are present; of the 144 winning sessions on them, 130 reproduce the win when replayed on the competition engine (114 frame by frame), and every game has at least two reproducible winning sessions. Median human actions per level track the competition's own baseline counts. For each level we count the sessions on the competition version that reached it (furthest level at least L − 1) and how many of those cleared it (furthest level at least L); levels with fewer than three such sessions are excluded from the medians. Only derived counts are published.

Think-aloud. A second public dataset (Liu et al., 2026; CC BY 4.0) records 144 human trajectories, thinking aloud, on the 25 public games; 132 replay to identity on the competition engine. It marks, per trajectory and game mechanism, the action at which a player first shows sufficient evidence of the mechanism; these labels are the dataset's own and partly machine-assisted. Two coders, blind to outcomes and to each other, mapped its mechanisms to the key insights of the 157 levels (kappa 0.80); a descriptive rule fixed before any outcome set the primaries; intervals come from a two-stage bootstrap over levels, then trajectories. Readings: at the levels the agent clears only when told the idea, a median of 58 actions to recognition against 23.5 at the levels it clears unchanged (difference 34.5, interval 6.25 to 55.0); at the 25 stuck levels no difference shown (83 against 31.75, interval −9.75 to 99.5), with only 13 of them recognised within the level and 74 of the 145 recognitions at an earlier level; of 107 intervals computed, 14 exclude zero against about 5.4 expected by chance.

The team's own play. Members of the team played the public games individually, by hand, and recorded routes, rules, insights and tricks; the routes are checked on the engine. In this paper these texts serve only as inputs to arms (the rule card, the learned writer's tuning data, the tricks of the hint entries, the law kinds) and the routes only as solution lengths.